



Simulating Flight Mission Activity

Activity: Demonstrate the ability to create and edit a flight mission using Ground Control Station (GCS) software by defining waypoints, setting flight parameters, and monitoring the mission in real-time.

Materials:

- Computer with drone simulation software (e.g., Gazebo, AirSim)
- GCS software compatible with the simulation
- Virtual drone model

Instructions:

1. Familiarise with GCS Software:

- Learn how to use the GCS software to create and manage flight missions.
- Understand the basic concepts of waypoints, flight parameters, and mission planning.

2. Create a Virtual Flight Mission:

- Use the GCS to define waypoints for the drone's flight path.
- Set flight parameters such as altitude, speed, and heading for each waypoint.
- Visualise the planned flight path in the simulation environment.

3. Execute the Mission:

- Load the flight mission into the GCS and start the simulation.
- Observe the drone's flight path and its adherence to the defined waypoints and parameters.
- Monitor the mission in real-time, making adjustments as needed.



4. Data Analysis:

- Record and analyse the flight data, including the drone's position, altitude, speed, and sensor readings.
- Evaluate the mission's success and identify areas for improvement.

5. Modifications and Re-execution:

- Modify the flight mission by adding, removing, or changing waypoints and parameters.
- Re-execute the mission and analyse the results.

6. Final Project:

- Create a comprehensive report or presentation summarising your findings and conclusions.
- Include the following:
 - A detailed description of the flight mission you created
 - Analysis of the drone's performance during the mission
 - Identification of any challenges or issues encountered